# Exercises

## Exercise 4

Define an RNN for regression, having 2 inputs, a hidden state of dimension 3, and 2 outputs. The activation function for the hidden state is ReLU. Write the equations for two time steps of forward pass through the network. For the inputs $\boldsymbol{X}_1 = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$, $\boldsymbol{X}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, if the weight matrices and bias vector are generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros), compute the outputs $\boldsymbol{O}_1$ and $\boldsymbol{O}_2$ of the network. Specify the total number of real parameters (weights and biases) for the layer.

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**Solution:**

- we have the following formulas:

$$\boldsymbol{H}_t = \phi(\boldsymbol{X}_t \boldsymbol{W}_{xh} + \boldsymbol{H}_{t-1} \boldsymbol{W}_{hh} + \boldsymbol{b}_h),$$

$$\boldsymbol{O}_t = \boldsymbol{H}_t \boldsymbol{W}_{hq} + \boldsymbol{b}_q,$$

where $\boldsymbol{X}_t \in \mathbb{R}^{n \times d}$ , $\boldsymbol{W}_{xh} \in \mathbb{R}^{d \times h}$, $\boldsymbol{W}_{hh} \in \mathbb{R}^{h \times h}$, $\boldsymbol{b}_h \in \mathbb{R}^{1 \times h}$, $\boldsymbol{H}_t \in \mathbb{R}^{n \times h}$ , $\boldsymbol{W}_{hq} \in \mathbb{R}^{h \times q}$, $\boldsymbol{b}_q \in \mathbb{R}^{1 \times q}$, $\boldsymbol{O}_t \in \mathbb{R}^{n \times q}$, and

$$\phi(x) = \text{ReLU}(x)$$

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- in our case, $n = 2$, $d = 2$, $h = 3$, $q = 2$
- we take

$$\boldsymbol{W}_{xh} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \in \mathbb{R}^{2\times 3},$$

$$\boldsymbol{W}_{hh} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} \in \mathbb{R}^{3\times 3},$$

$$\boldsymbol{b}_h = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \in \mathbb{R}^{1\times 3},$$

$$\boldsymbol{W}_{hq} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 2},$$

$$\boldsymbol{b}_q = \begin{bmatrix} 1 & -1 \end{bmatrix} \in \mathbb{R}^{1\times 2},$$

$$\boldsymbol{H}_0 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{2\times 3}.$$

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- now, we apply the formulas:

$$
\begin{aligned}
\boldsymbol{H}_1 &= \phi\left(\begin{bmatrix}1 & 0\\ 1 & -1\end{bmatrix}\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix} + \begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}1 & 0 & -1\\ 0 & 1 & 0\\ 1 & -1 & 0\end{bmatrix} + \begin{bmatrix}1 & 0 & -1\end{bmatrix}\right)\\
&= \phi\left(\begin{bmatrix}1 & 0 & 1\\ 1 & -1 & 1\end{bmatrix} + \begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix} + \begin{bmatrix}1 & 0 & -1\\ 1 & 0 & -1\end{bmatrix}\right)\\
&= \phi\left(\begin{bmatrix}2 & 0 & 0\\ 2 & -1 & 0\end{bmatrix}\right)\\
&= \begin{bmatrix}2 & 0 & 0\\ 2 & 0 & 0\end{bmatrix} \in \mathbb{R}^{2\times 3},
\end{aligned}
$$

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$$
\begin{aligned}
\boldsymbol{H}_2 &= \phi\left(\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}+\begin{bmatrix}2 & 0 & 0\\ 2 & 0 & 0\end{bmatrix}\begin{bmatrix}1 & 0 & -1\\ 0 & 1 & 0\\ 1 & -1 & 0\end{bmatrix}+\begin{bmatrix}1 & 0 & -1\end{bmatrix}\right)\\
&= \phi\left(\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 1\end{bmatrix}+\begin{bmatrix}2 & 0 & -2\\ 2 & 0 & -2\end{bmatrix}+\begin{bmatrix}1 & 0 & -1\\ 1 & 0 & -1\end{bmatrix}\right)\\
&= \phi\left(\begin{bmatrix}3 & 1 & -3\\ 4 & 0 & -2\end{bmatrix}\right)\\
&= \begin{bmatrix}3 & 1 & 0\\ 4 & 0 & 0\end{bmatrix}\in\mathbb{R}^{2\times 3},
\end{aligned}
$$

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$$
\begin{aligned}
\boldsymbol{O}_1 &= \begin{bmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 3 & -1 \\ 3 & -1 \end{bmatrix} \in \mathbb{R}^{2\times 2},
\end{aligned}
$$

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$$
\begin{aligned}
\boldsymbol{O}_2 &= \begin{bmatrix} 3 & 1 & 0 \\ 4 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 3 & 1 \\ 4 & 0 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 4 & 0 \\ 5 & -1 \end{bmatrix} \in \mathbb{R}^{2\times 2},
\end{aligned}
$$

$$
\begin{aligned}
\text{Total parameters} &= 2 \times 3 + 3 \times 3 + 1 \times 3 + 3 \times 2 + 1 \times 2 \\
&= 6 + 9 + 3 + 6 + 2 \\
&= 26.
\end{aligned}
$$